

umble

Project Report

1. Readme

This is from Runoffgrove Umble, and is based on that schematic.

This pedal is verified. It is very loud. You must adjust the biases to around 4.5v on pin3 of each trimmer.

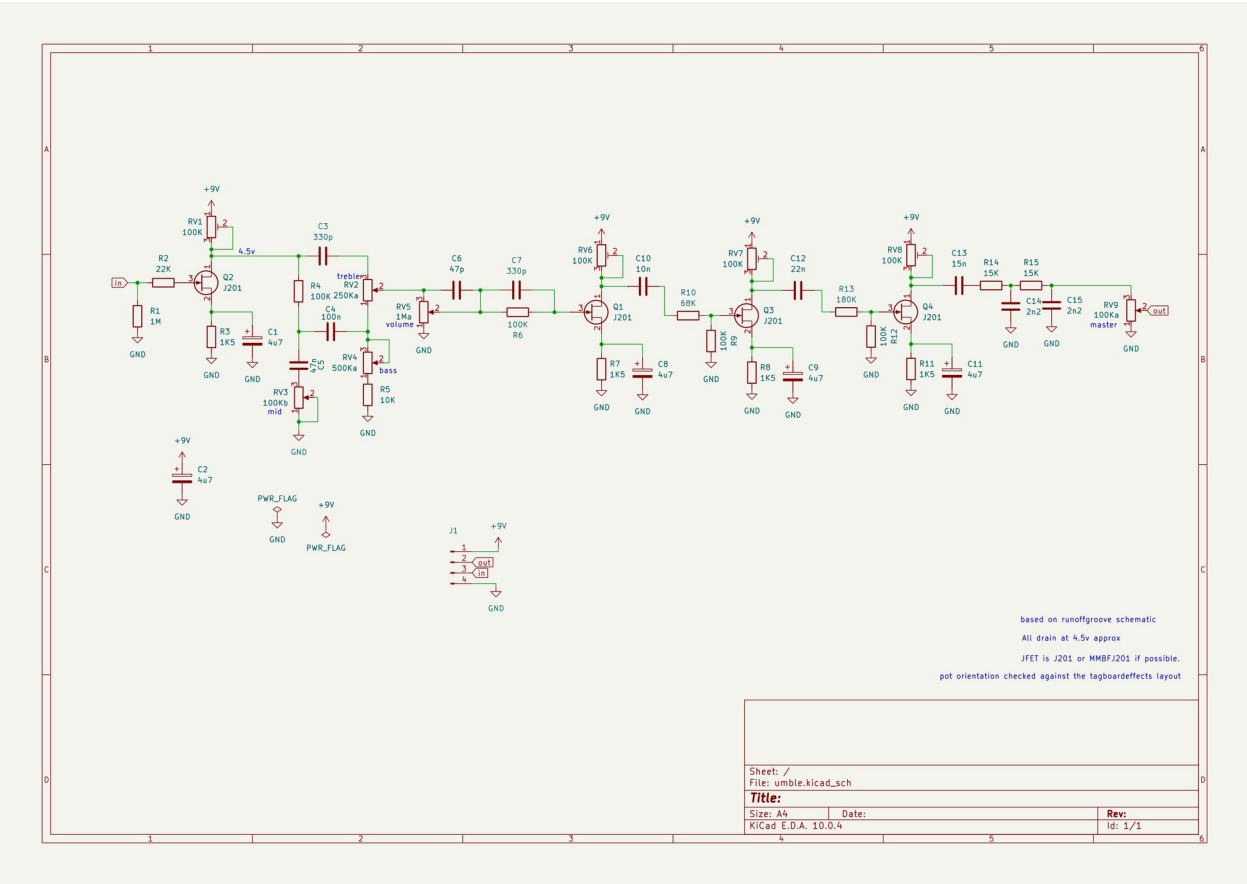
Out of an enclosure it has a lot of noise, which I think is common for higher-gain pedals. All 5 controls seem to work the way that they are intended.

This only has footprints for SMD because I did not think there was enough room for a through-hole and SMD. There probably was enough space to include a bigger footprint to include through-hole. Oh well. I soldered on the J201 first, and then used other pads on the empty pcb to test that the soldered J201s were functional. All 4 tested out as NJFET with a peak tester.

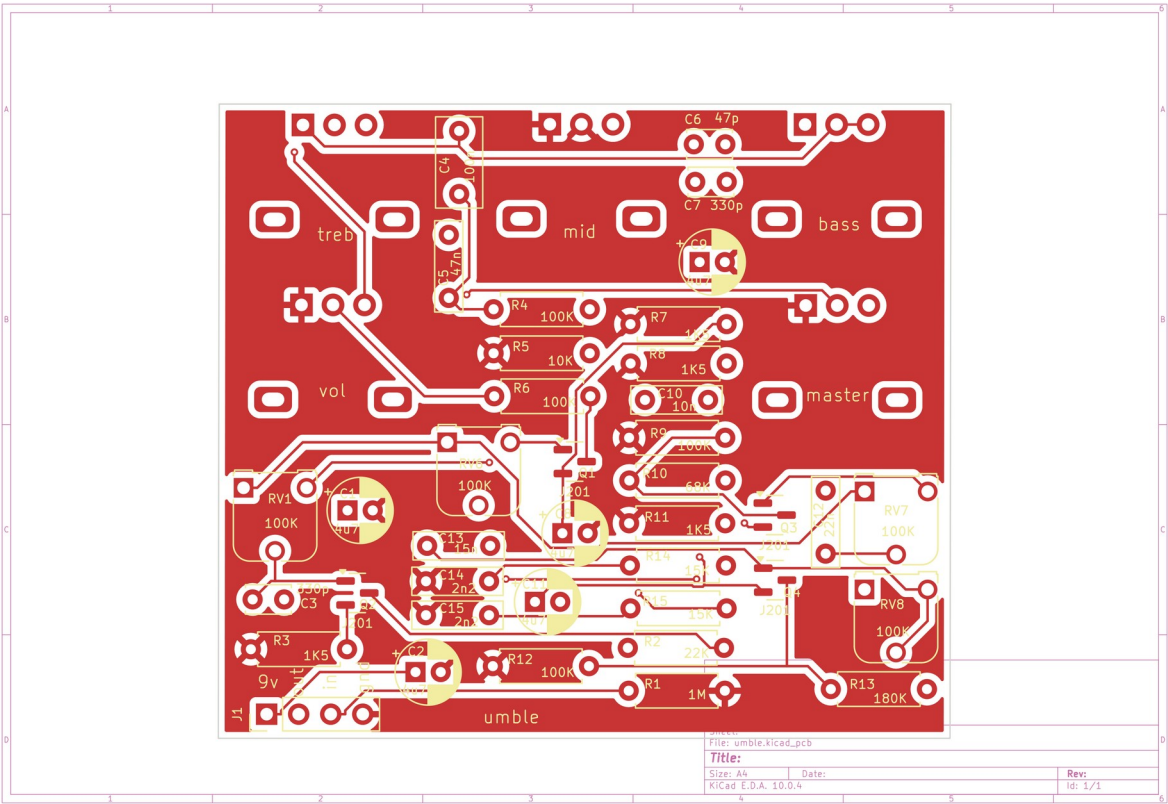
Sourcing all the pots with a RG097 footprint was a bit of a PITA, but I do love how tight these pots are when it is all boxed up.

Not boxed yet, tho.

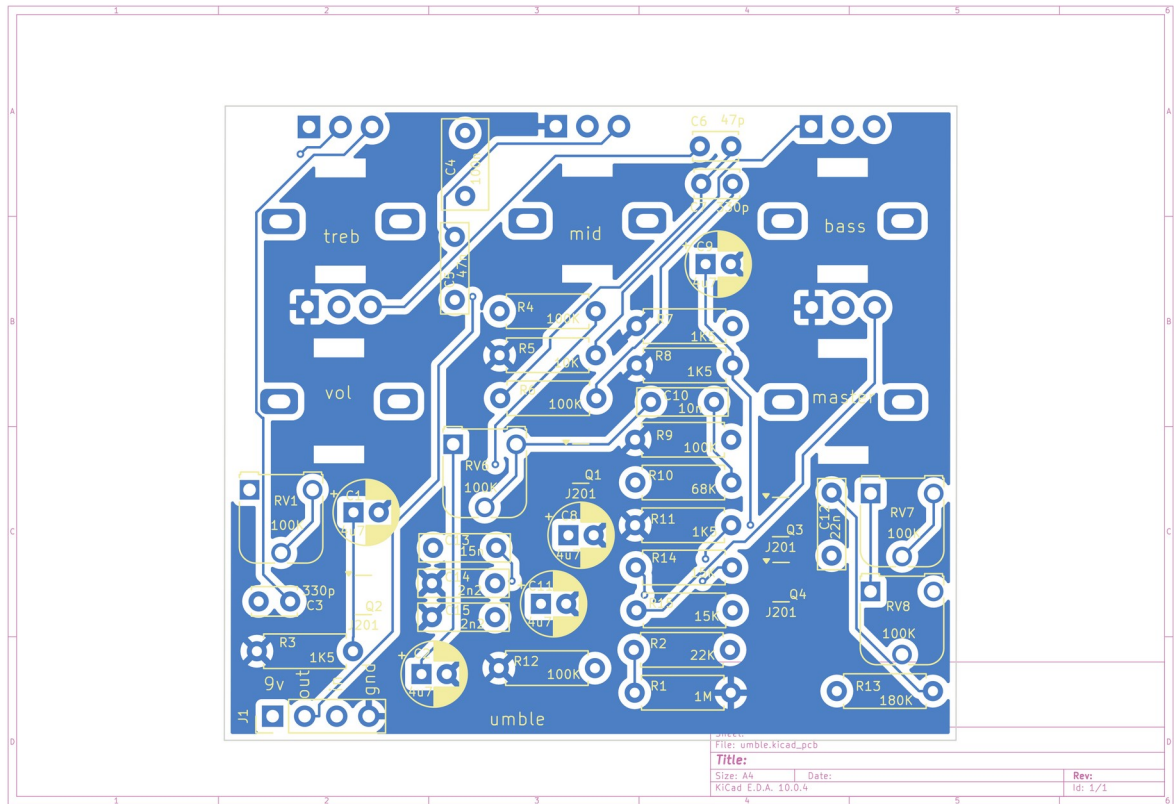
2. Schematic



3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C2,C8,C9,C11	4u7	Capacitor_THT :CP_Radial_D5.0mm_P2.00mm	5	
C3,C7	330p	Capacitor_THT :C_Disc_D3.4mm_W2.1mm_P2.50mm	2	
C4	100n	Capacitor_THT :C_Rect_L7.0mm_W3.5mm_P5.00mm	1	
C5	47n	Capacitor_THT :C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C6	47p	Capacitor_THT :C_Disc_D3.4mm_W2.1mm_P2.50mm	1	
C10	10n	Capacitor_THT :C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C12	22n	Capacitor_THT :C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C13	15n	Capacitor_THT :C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C14,C15	2n2	Capacitor_THT :C_Rect_L7.0mm_W2.0mm_P5.00mm	2	
J1	Conn_01x04_Pin	Connector_PinHeader_2.54mm:PinHeader_	1	

		1x04_P2.54mm_Vertical		
Q1-Q4	J201	Package_TO_SOT_SMD:SOT-23	4	
R1	1M	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R2	22K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R3,R7,R8,R11	1K5	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	4	
R4,R6,R9,R12	100K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	4	
R5	10K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R10	68K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R13	180K	Resistor_THT:	1	

		R_Axial_DIN02 07_L6.3mm_D 2.5mm_P7.62 mm_Horizont al		
R14,R15	15K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P7.62 mm_Horizont al	2	
RV1,RV6-RV8	100K	Potentiometer _THT:Potentio meter_Runtr n_RM- 065_Vertical	4	
RV2	250Ka	Potentiometer _THT:Potentio meter_Alps_R K09L_Single_V ertical	1	
RV3	100Kb	Potentiometer _THT:Potentio meter_Alps_R K09L_Single_V ertical	1	
RV4	500Ka	Potentiometer _THT:Potentio meter_Alps_R K09L_Single_V ertical	1	
RV5	1Ma	Potentiometer _THT:Potentio meter_Alps_R K09L_Single_V ertical	1	
RV9	100Ka	Potentiometer _THT:Potentio meter_Alps_R K09L_Single_V ertical	1	

Total line items: 25

pics:

